Matthew Leung

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Technical Profile

Computer engineering M.S. candidate focused on the intersection of low-level software, embedded systems, hardware, and cybersecurity. Hands-on experience spanning UAV penetration testing, firmware and wireless security, Linux/OTA infrastructure, real-time systems, computer architecture, and hardware/software integration.

Education

University of California, San Diego **GPA: 3.98**

Master of Science in Computer Engineering *Expected June 2027*

Bachelor of Science in Computer Engineering *September 2022 – June 2026*

- Selected for UC San Diego's accelerated B.S./M.S. program based on academic performance during undergraduate study
- Member of Tau Beta Pi Engineering Honor Society

Experience

Electrical Engineering & Cybersecurity Intern June 2026 – Present

TrustThink

- Perform penetration testing on unmanned aerial systems across firmware, wireless interfaces, and physical ports
- Identified, reproduced, and documented 5+ high- and medium-severity vulnerabilities, including unsigned protocols, unencrypted filesystems, and exposed storage and communication ports
- Develop and execute MAVLink message-spoofing scripts to exploit firmware vulnerabilities and evaluate potential security impact on mission-critical systems
- Conduct software-in-the-loop testing of flight controller failsafes and assess Linux-based companion computer behavior under network attacks

Computer Science Tutor (CSE 30) January 2026 – March 2026

University of California, San Diego

- Led weekly discussion sections for 15+ students, translating computer architecture concepts, including caching, stack frames, and ARM calling conventions, into practical examples and debugging exercises
- Mentored and collaborated with 3-5 students during office hours, providing hands-on GDB debugging support and helping develop systematic approaches to diagnosing software and architecture-level problems
- Developed and refined course-aligned instructional materials to reinforce lecture concepts and strengthen students' application of low-level programming and computer architecture principles

Projects

Junkyard Computing – OTA Updates | *Linux, Bash, dracut, btrfs* March 2026 – June 2026

- Contributed to a Google-sponsored research project developing scalable compute clusters from repurposed Google Pixel devices, focusing on OTA update infrastructure
- Engineered an A/B rootfs update system using btrfs subvolumes to safely deploy and roll back OS updates
- Automated rootfs versioning and boot-time update management in a custom dracut module
- Designed an update staging system that moves rootfs storage from the 8 GB super partition to userdata, expanding available storage by up to 25x

RealTime Display | *ESP32, C++, FreeRTOS, Python, Flask, Onshape* September 2025 – December 2025

- Designed and built an integrated ESP32-based LED platform combining clock, weather, Spotify visualization, and live sports tracking into a single real-time system
- Implemented FreeRTOS mutexes and task synchronization to coordinate concurrent networking, rendering, input handling, and display updates
- Developed a modular client-server architecture for custom widgets and integrated the electronics with a fabricated 3D-printed enclosure and physical controls

Lava LiDAR Game | *Arduino, Python, C++* April 2025 – June 2025

- Integrated a LiDAR sensor with an Arduino Uno over I2C to enable real-time player tracking and interactive gameplay
- Developed a serial communication pipeline connecting the embedded controller with the Python game engine and coordinated data flow between hardware and software components
- Presented the completed system at UCSD's IEEE Quarterly Projects Showcase, earning 1st Place — Spring 2025

Technical Skills

Low-Level / Systems: C/C++, Python, Linux, Bash, GDB, Computer Architecture, SystemVerilog
Embedded / Real-Time: ESP32, Arduino, FreeRTOS, Digital Systems Design
Cybersecurity: Penetration Testing, Firmware Security, MAVLink, Wireshark, Ghidra
Hardware / EDA: LTspice, ModelSim, Cadence Virtuoso, UART, CAN, I2C
Software / Cloud: Git, Java, TypeScript, JavaScript, React, Node.js, Flask, JUnit, Google Cloud Platform, Hugging Face